A01 · The Sort That Stole Everyone's Grade — Practice A2 · 9618 §13.1

Name: _____ Due: _____

Every declaration on this sheet is marked in **halves** — the `TYPE` line scores separately from what follows it. **So write the half you know.** A blank line scores nothing; half a line scores half.

1. A city bicycle-hire scheme records every hire. [10]

The bikes available are of these models only: `Classic, Electric, Cargo, Folding, Tandem`.
Each hire records:

- a hire number, which may be any combination of letters and digits
- the customer's name
- the date of hire
- the model of bike used
- whether the bike has been returned

(a) Write pseudocode to declare the **enumerated** data type `BikeModel`. [2]

(b) Write pseudocode to declare the **composite** data type `Hire`. Use the enumerated type from part (a). [4]

(c) Write the pseudocode statement to set up a variable for **one** record of type `Hire`. [1]

(d) Write pseudocode to store the following hire in the variable you set up in part (c). [3]

Field	Data
hire number	`BH4417`
customer name	`R Okafor`
date of hire	`03/09/2026`
model of bike	`Electric`
returned	`FALSE`

2. Pointers. Write pseudocode to declare each of the following pointer data types. [2]

(a) `TCustomerPointer`, which points at a `STRING`

(b) `TModelPointer`, which points at the enumerated type you declared in question 1(a)

3. A set. A theatre labels its seat rows with the capital letters `A` to `E` only. [4]

Write pseudocode to declare a set data type `RowSet`, and to define a set `ValidRows` containing those five letters.

4. Describe what is meant by a **non-composite** data type. Include an example in your answer. [3]

5. Describe a **set** data type. [3]

6. A student stores a club's members using two arrays: [2]

```
DECLARE MemberName : ARRAY[1:200] OF STRING
DECLARE MemberFee  : ARRAY[1:200] OF REAL
```

The program then sorts **`MemberFee`** into ascending order. It runs without any error message.
Explain what is now wrong with the data, and state the **user-defined data type** that would have prevented it.

🍦 **The Ice-Cream Challenge — records, rebuilt so they cannot lie.** (0 marks. 1 ice cream.) Open **`challenge_A01_Records.py`**, fill in **`build_class()`** and **`sort_by_grade()`**, and run it until every row is a ✓. **It cannot be faked:** the judge checks that the set of (name, grade) *pairs* is identical before and after — so if Chen's 58 ends up on Amir you fail, however well sorted the numbers are.

Want to dig? In question 1(a) you refused to store the bike model as a number, because a number lets somebody write `7` and mean nothing. Now find out how an enumerated type is *actually stored*. The compiler numbers your values in the order you wrote them — `Classic` is 0, `Electric` is 1, and so on — so underneath it **is** an integer after all. So was the argument for nothing? No — and working out what the enumerated type still buys you, when the storage is identical, is the real lesson. *(Clue: the difference is not in what the machine holds, but in what the compiler will let you type.)*

Additional space. If you need more room for any answer, continue here and write the question number in the margin beside it.